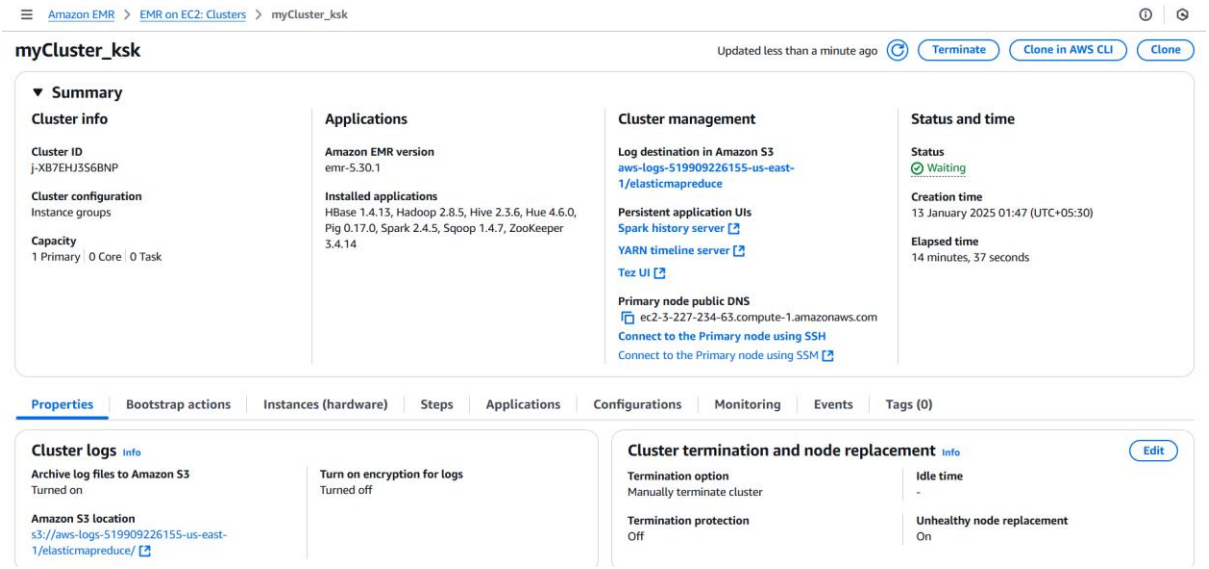
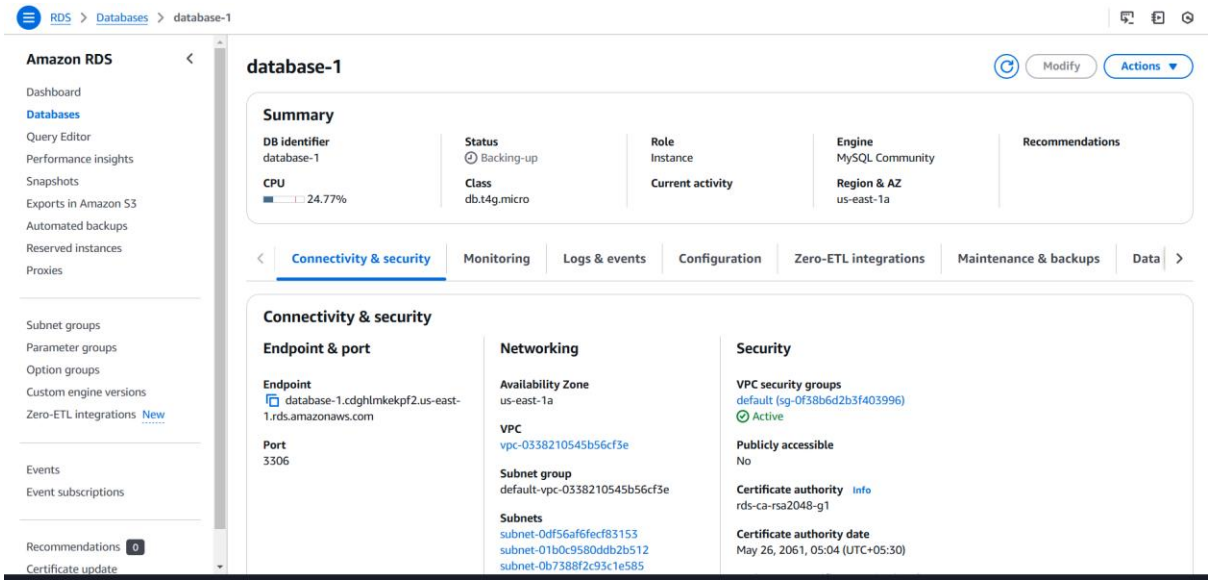
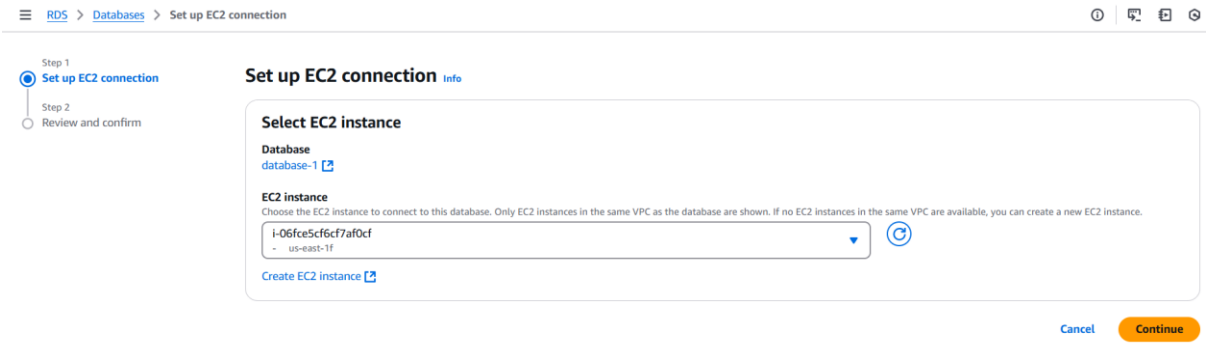
Step 1 – Create EMR instance in AWS with applications such as Hadoop, Hive, Sqoop etc installed.



Step 2 – Create RDS instance in AWS

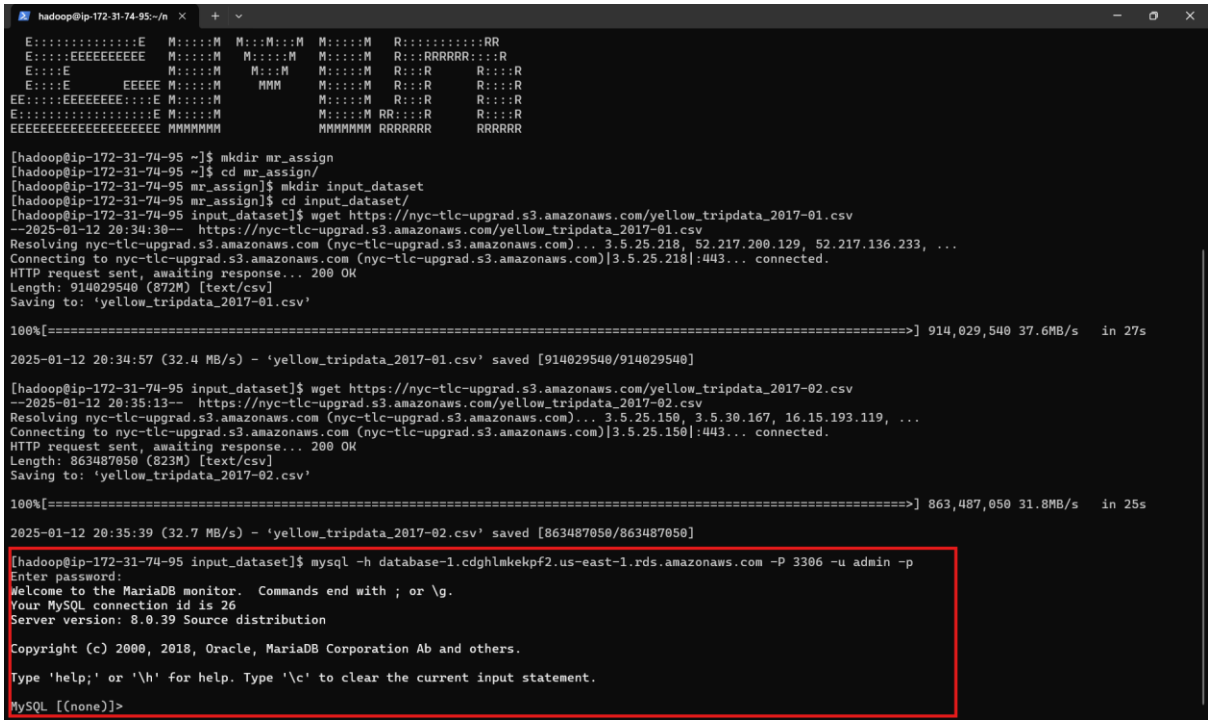


Step 3 – Connect RDS instance with EMR instance



Step 4 – Logging into RDS through EMR instance using the command

mysql -h database-1.cdghlmkekpf2.us-east-1.rds.amazonaws.com -P 3306 -u admin -p



```
hadoop@ip-172-31-74-95:~/n

  E::::::::::::::E   M::::::M  M::::M::::M  M::::::M   R::::::::::::RR
  E::::::EEEEEEEEEE   M::::::M   M::::::M   M::::::M   R::::RRRRRR::::R
  E::::E              M::::::M    M::::M    M::::::M   R::::R      R::::R
  E::::E       EEEEE  M::::::M     MMM      M::::::M   R::::R      R::::R
EE::::::EEEEEEEE::::E M::::::M              M::::::M   R::::R      R::::R
E::::::::::::::::::E  M::::::M              M::::::M RR::::R      R::::R
EEEEEEEEEEEEEEEEEEEEE MMMMMMMM              MMMMMMM RRRRRRR      RRRRRR

[hadoop@ip-172-31-74-95 ~]$ mkdir mr_assign
[hadoop@ip-172-31-74-95 ~]$ cd mr_assign/
[hadoop@ip-172-31-74-95 mr_assign]$ mkdir input_dataset
[hadoop@ip-172-31-74-95 mr_assign]$ cd input_dataset/
[hadoop@ip-172-31-74-95 input_dataset]$ wget https://nyc-tlc-upgrad.s3.amazonaws.com/yellow_tripdata_2017-01.csv
--2025-01-12 20:34:30--  https://nyc-tlc-upgrad.s3.amazonaws.com/yellow_tripdata_2017-01.csv
Resolving nyc-tlc-upgrad.s3.amazonaws.com (nyc-tlc-upgrad.s3.amazonaws.com)... 3.5.25.218, 52.217.200.129, 52.217.136.233, ...
Connecting to nyc-tlc-upgrad.s3.amazonaws.com (nyc-tlc-upgrad.s3.amazonaws.com)|3.5.25.218|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 914029540 (872M) [text/csv]
Saving to: 'yellow_tripdata_2017-01.csv'

100%[=================================================================================================>] 914,029,540 37.6MB/s   in 27s

2025-01-12 20:34:57 (32.4 MB/s) - 'yellow_tripdata_2017-01.csv' saved [914029540/914029540]

[hadoop@ip-172-31-74-95 input_dataset]$ wget https://nyc-tlc-upgrad.s3.amazonaws.com/yellow_tripdata_2017-02.csv
--2025-01-12 20:35:13--  https://nyc-tlc-upgrad.s3.amazonaws.com/yellow_tripdata_2017-02.csv
Resolving nyc-tlc-upgrad.s3.amazonaws.com (nyc-tlc-upgrad.s3.amazonaws.com)... 3.5.25.150, 3.5.30.167, 16.15.193.119, ...
Connecting to nyc-tlc-upgrad.s3.amazonaws.com (nyc-tlc-upgrad.s3.amazonaws.com)|3.5.25.150|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 863487050 (823M) [text/csv]
Saving to: 'yellow_tripdata_2017-02.csv'

100%[=================================================================================================>] 863,487,050 31.8MB/s   in 25s

2025-01-12 20:35:39 (32.7 MB/s) - 'yellow_tripdata_2017-02.csv' saved [863487050/863487050]

[hadoop@ip-172-31-74-95 input_dataset]$ mysql -h database-1.cdghlmkekpf2.us-east-1.rds.amazonaws.com -P 3306 -u admin -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MySQL connection id is 26
Server version: 8.0.39 Source distribution

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MySQL [(none)]>
```

Step 5 – Creating data base '**upgrad_assign**' and table '**yellow_taxi**'

create database upgrad_assign;

use upgrad_assign;

create table yellow_taxi (VendorID int, tpep_pickup_datetime datetime, tpep_dropoff_datetime datetime, passenger_count int, trip_distance float, RatecodeID int, store_and_fwd_flag varchar(10), PULocationID int, DOLocationID int, payment_type int, fare_amount float, extra float, mta_tax float, tip_amount float, tolls_amount float, improvement_surcharge float, total_amount float, congestion_surcharge float, airport_fee float);

```
hadoop@ip-172-31-74-95:~/n

2025-01-12 20:34:57 (32.4 MB/s) - 'yellow_tripdata_2017-01.csv' saved [914029540/914029540]

[hadoop@ip-172-31-74-95 input_dataset]$ wget https://nyc-tlc-upgrad.s3.amazonaws.com/yellow_tripdata_2017-02.csv
--2025-01-12 20:35:13--  https://nyc-tlc-upgrad.s3.amazonaws.com/yellow_tripdata_2017-02.csv
Resolving nyc-tlc-upgrad.s3.amazonaws.com (nyc-tlc-upgrad.s3.amazonaws.com)... 3.5.25.150, 3.5.30.167, 16.15.193.119, ...
Connecting to nyc-tlc-upgrad.s3.amazonaws.com (nyc-tlc-upgrad.s3.amazonaws.com)|3.5.25.150|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 863487050 (823M) [text/csv]
Saving to: 'yellow_tripdata_2017-02.csv'

100%[===================================================================================================>] 863,487,050 31.8MB/s   in 25s

2025-01-12 20:35:39 (32.7 MB/s) - 'yellow_tripdata_2017-02.csv' saved [863487050/863487050]

[hadoop@ip-172-31-74-95 input_dataset]$ mysql -h database-1.cdghlmkekpf2.us-east-1.rds.amazonaws.com -P 3306 -u admin -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MySQL connection id is 26
Server version: 8.0.39 Source distribution

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MySQL [(none)]> show databases;
+--------------------+
| Database           |
+--------------------+
| information_schema |
| mysql              |
| performance_schema |
| sys                |
+--------------------+
4 rows in set (0.00 sec)

MySQL [(none)]> create database upgrad_assign;
Query OK, 1 row affected (0.02 sec)

MySQL [(none)]> use upgrad_assign;
Database changed
MySQL [upgrad_assign]> create table yellow_taxi (VendorID int, tpep_pickup_datetime datetime, tpep_dropoff_datetime datetime, passenger_count int, trip_dist
ance float, RatecodeID int, store_and_fwd_flag varchar(10), PULocationID int, DOLocationID int, payment_type int, fare_amount float, extra float, mta_tax fl
oat, tip_amount float, tolls_amount float, improvement_surcharge float, total_amount float, congestion_surcharge float, airport_fee float);
Query OK, 0 rows affected (0.04 sec)

MySQL [upgrad_assign]>
```

<u>Step 6</u> – Loading the data into MySQL table.

The csv files are already downloaded to EMR cluster into the folder '**/home/hadoop/mr_assign/input_dataset/**'.

Using the following commands, **yellow_tripdata_2017-01.csv** and **yellow_tripdata_2017-02.csv** will be loaded to MySQL table '**yellow_taxi**'

load data local infile '/home/hadoop/mr_assign/input_dataset/yellow_tripdata_2017-01.csv' into table yellow_taxi fields terminated by ',' lines terminated by '\n' ignore 1 lines;

load data local infile '/home/hadoop/mr_assign/input_dataset/yellow_tripdata_2017-02.csv' into table yellow_taxi fields terminated by ',' lines terminated by '\n' ignore 1 lines;

```
hadoop@ip-172-31-74-95:~/n

100%[===================================================================================================================>] 863,487,050 31.8MB/s   in 25s

2025-01-12 20:35:39 (32.7 MB/s) - 'yellow_tripdata_2017-02.csv' saved [863487050/863487050]

[hadoop@ip-172-31-74-95 input_dataset]$ mysql -h database-1.cdghlmkekpf2.us-east-1.rds.amazonaws.com -P 3306 -u admin -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MySQL connection id is 26
Server version: 8.0.39 Source distribution

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MySQL [(none)]> show databases;
+--------------------+
| Database           |
+--------------------+
| information_schema |
| mysql              |
| performance_schema |
| sys                |
+--------------------+
4 rows in set (0.00 sec)

MySQL [(none)]> create database upgrad_assign;
Query OK, 1 row affected (0.02 sec)

MySQL [(none)]> use upgrad_assign;
Database changed
MySQL [upgrad_assign]> create table yellow_taxi (VendorID int, tpep_pickup_datetime datetime, tpep_dropoff_datetime datetime, passenger_count int, trip_dist
ance float, RatecodeID int, store_and_fwd_flag varchar(10), PULocationID int, DOLocationID int, payment_type int, fare_amount float, extra float, mta_tax fl
oat, tip_amount float, tolls_amount float, improvement_surcharge float, total_amount float, congestion_surcharge float, airport_fee float);
Query OK, 0 rows affected (0.04 sec)

MySQL [upgrad_assign]> load data local infile '/home/hadoop/mr_assign/input_dataset/yellow_tripdata_2017-01.csv' into table yellow_taxi fields terminated by
 ',' lines terminated by '\n' ignore 1 lines;
Query OK, 9710820 rows affected, 65535 warnings (2 min 2.47 sec)
Records: 9710820  Deleted: 0  Skipped: 0  Warnings: 19421640

MySQL [upgrad_assign]> load data local infile '/home/hadoop/mr_assign/input_dataset/yellow_tripdata_2017-02.csv' into table yellow_taxi fields terminated by
 ',' lines terminated by '\n' ignore 1 lines;
Query OK, 9169775 rows affected, 65535 warnings (2 min 5.92 sec)
Records: 9169775  Deleted: 0  Skipped: 0  Warnings: 18339550

MySQL [upgrad_assign]>
```

select count(*) from yellow_taxi;

```
hadoop@ip-172-31-74-95:~/n

Your MySQL connection id is 26
Server version: 8.0.39 Source distribution

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MySQL [(none)]> show databases;
+--------------------+
| Database           |
+--------------------+
| information_schema |
| mysql              |
| performance_schema |
| sys                |
+--------------------+
4 rows in set (0.00 sec)

MySQL [(none)]> create database upgrad_assign;
Query OK, 1 row affected (0.02 sec)

MySQL [(none)]> use upgrad_assign;
Database changed
MySQL [upgrad_assign]> create table yellow_taxi (VendorID int, tpep_pickup_datetime datetime, tpep_dropoff_datetime datetime, passenger_count int, trip_dist
ance float, RatecodeID int, store_and_fwd_flag varchar(10), PULocationID int, DOLocationID int, payment_type int, fare_amount float, extra float, mta_tax fl
oat, tip_amount float, tolls_amount float, improvement_surcharge float, total_amount float, congestion_surcharge float, airport_fee float);
Query OK, 0 rows affected (0.04 sec)

MySQL [upgrad_assign]> load data local infile '/home/hadoop/mr_assign/input_dataset/yellow_tripdata_2017-01.csv' into table yellow_taxi fields terminated by
 ',' lines terminated by '\n' ignore 1 lines;
Query OK, 9710820 rows affected, 65535 warnings (2 min 2.47 sec)
Records: 9710820  Deleted: 0  Skipped: 0  Warnings: 19421640

MySQL [upgrad_assign]> load data local infile '/home/hadoop/mr_assign/input_dataset/yellow_tripdata_2017-02.csv' into table yellow_taxi fields terminated by
 ',' lines terminated by '\n' ignore 1 lines;
Query OK, 9169775 rows affected, 65535 warnings (2 min 5.92 sec)
Records: 9169775  Deleted: 0  Skipped: 0  Warnings: 18339550

MySQL [upgrad_assign]> select count(*) from yellow_taxi;
+----------+
| count(*) |
+----------+
| 18880595 |
+----------+
1 row in set (44.85 sec)

MySQL [upgrad_assign]>
```